NightEcho — Full Project Proposal

Oct 31, 2026 · @amanda

NightEcho turns the phone on a nightstand into a contactless sonar sleep monitor that screens for REM sleep behavior disorder (RBD), the strongest early warning sign of Parkinson's disease and Lewy body dementia, and supports dementia and autism sleep care. This proposal lays out every step from today to the Innovation Stage deadline (Jan 7, 2027) and the Innovation Summit (Apr 21, 2027).

At a glance

NightEcho needs no hardware beyond a phone, costs under $500 to prototype, and can show a working demo plus simulated results by January 7.

Item	Plan
Title	NightEcho: A Contactless Smartphone Sonar Platform and Modular Machine Learning Engine for Early Screening of REM Sleep Behavior Disorder, Neurodegenerative Risk, and Pediatric Autism Sleep Disturbances
Problem	RBD is the strongest early sign of Parkinson's and Lewy body dementia, but diagnosis needs an overnight video sleep study. Dementia caregivers sleep through night wandering. Many autistic children refuse wearable trackers.
Solution	An app that uses the phone's own speaker and microphone as sonar to track breathing, movement, bed exits and vocal events, feeding three modules: RBD screening, dementia alerts, autism sleep tracking.
Who it serves	Adults 50+ who act out dreams; dementia caregivers and memory-care facilities; parents of autistic children; sleep clinics; Parkinson's trial sponsors

Item	Plan
What is new	RBD screening with only a phone's speaker and microphone; gates that remove known RBD lookalikes; a simulator that turns public sleep-lab recordings into training data for sonar
Key targets	RBD AUC > 0.85; breathing-rate error < 1 breath/min; bed-exit recall > 95%; caregiver alert < 10 s
Category	Health and Nutrition
Budget	Under $500 for the prototype, plus the $499 Innovation Stage fee (financial aid available)
Deadlines	Innovation Stage: Jan 7, 2027, 11:59 PM CT. Finalists announced: February 2027. Innovation Summit: Apr 21–24, 2027, Space Center Houston

Problem and significance

The warning signs for all three groups happen at night, and today's tools need either a sleep lab or something worn on the body.

RBD: a warning sign years ahead

- People with RBD act out their dreams by shouting, punching or kicking, because the normal muscle paralysis of REM sleep is lost.
- In a 1,280-patient multicenter study, people with isolated RBD developed Parkinson's disease, Lewy body dementia or multiple system atrophy at 6.3% per year; 73.5% had converted after 12 years (Postuma et al., Brain 2019).
- In the HypnoLaus general-population study, 21 of 1,997 adults (1.06%) met sleep-study criteria for RBD (Haba-Rubio et al., SLEEP). Most people with RBD are never diagnosed.
- Diagnosis requires video-polysomnography with expert scoring, which does not scale to screening or to finding patients for trials (Skjærbæk et al., 2026).
- Questionnaires are cheap but flag many false positives from lookalikes such as sleep apnea, nightmares, medications and NREM parasomnias (Cesari et al., J Sleep Res 2025).

- Neuroprotective drug trials need 142–366 RBD patients per study arm, so finding these people early is a bottleneck for research (Postuma et al., 2019).

Dementia: nights drive injuries and moves to nursing homes

- An estimated 7.4 million Americans age 65 and older live with Alzheimer's dementia in 2026 (Alzheimer's Association).
- Nighttime activity raises the risk of injury and unattended exits from the home, and in one caregiver study it frequently led to nursing-home placement (Rowe et al., Alzheimer's & Dementia 2009).
- Fragmented sleep signals risk: among 737 older adults, those in the top 10% for sleep fragmentation had 1.5 times the risk of developing Alzheimer's compared with the bottom 10% (Lim et al., Sleep 2013).

Autism: sleep problems are common and hard to measure

- About 1 in 31 U.S. 8-year-olds has an autism diagnosis (CDC ADDM Network, 2022 data, released April 2025).
- Up to 81% of autistic children have sleep problems. When 214 autistic 2- to 6-year-olds were asked to wear an activity tracker for 14 days, device intolerance cost 42% of the cohort's data (Rapp et al., Sleep 2025).

The gap: no current tool screens for RBD, alerts dementia caregivers and tracks children's sleep without something to wear or buy.

Prior work and novelty

Phone sonar for sleep is not new: SleepScore has used it since 2018. NightEcho's novelty is what the sonar is used for (RBD screening, dementia alerts, children's sleep) and how the model is trained before any patient data exists.

Approach	Examples	Limitation for NightEcho's users
Video sleep study	Hospital sleep lab (the diagnostic gold standard)	One night, costly, needs expert scoring
Questionnaires	RBD single-question screen	Many false positives from lookalikes (Cesari et al., 2025)

Approach	Examples	Limitation for NightEcho's users
Body-worn sensors	Wrist actigraphy (ActiTect), lower-back sensor, self-applied electrodes	Must be worn every night; many autistic children refuse
Bed and radar sensors	Under-mattress sensor in 83 people with Alzheimer's; MinderCare radar in 30 dementia homes	Extra hardware in every home; none yet validated for RBD
Phone sonar	UW ApneaApp, licensed into the SleepScore app (ResMed joint venture, 2018); sonar sleep-apnea screener, 94% sensitivity, 97% specificity	Built for general sleep and apnea, not RBD, dementia alerts or children

What NightEcho adds

1. **RBD screening from the phone's own speaker and microphone**, with no wearable, camera or bed sensor. Our searches found no published or commercial example; the novelty check below must confirm this.
2. **Lookalike gates.** A REM gate, a breathing-pause check and a range gate each remove a specific RBD mimic that defeats questionnaires.
3. **A PSG-to-sonar simulator.** Public sleep-lab recordings with confirmed diagnoses become sonar-like training data, so the model learns RBD before any patient sleeps beside a phone.
4. **One sensing engine, three decision modules** for RBD, dementia and autism.
5. **Privacy by design.** Raw audio never leaves the phone's memory and is never saved.

Novelty check (do this in week 1)

- [ ] Search Google Patents for: "smartphone sonar REM sleep behavior disorder", "acoustic dream enactment detection", "contactless RBD screening", "FMCW audio sleep movement".
- [ ] Run the same searches on Google Scholar and PubMed, adding "phone", "acoustic" and "ultrasound".
- [ ] Search the App Store and Google Play for "RBD", "dream enactment" and "Parkinson's sleep".

- [ ] Read the claims of the 3–5 closest patents (start with ResMed's and SleepScore's sonar patents) and write one sentence on how NightEcho differs from each.
- [ ] Log every search (date, query, top results, our difference) in a prior-art log. Conrad's scoring guide asks whether an online search confirmed the idea is unique.
- [ ] If you want a patent, talk to your coach about a provisional filing before the website and video go public in January. U.S. law gives inventors one year after their own disclosure; most other countries give none.

Objectives, hypotheses and success criteria

The primary outcome is O3: subject-level AUC for RBD on held-out subjects. Every other objective is secondary, and each has a pass/fail target fixed before testing.

#	Objective	Hypothesis	Test	Pass if
O1	Build the phone sonar engine	Phone sonar measures breathing as well as a chest belt	10 adults × 2 home nights, phone vs Vernier respiration belt	Mean absolute breathing-rate error < 1 breath/min
O2	Build the PSG-to-sonar simulator	The signal chain recovers what was simulated	Simulated nights from CAP	Breathing correlation $r \geq 0.9$; ≥ 90% of injected limb movements found
O3	Screen for RBD (primary)	Sonar-type signals separate RBD from non-RBD subjects	Leave-one-subject-out on 103 CAP subjects	AUC > 0.85, with a 95% confidence interval reported
O4	Remove RBD lookalikes	The gates cut false positives in mimic disorders	Same model with and without gates, on the 63 CAP mimic subjects	Fewer false positives with gates (McNemar's test, $p < 0.05$)
O5	Alert dementia caregivers	Bed exits are caught quickly and reliably	Scripted bedroom trials	Recall > 95%; median alert < 10 s

#	Objective	Hypothesis	Test	Pass if
O6	Track children's sleep acceptably (exploratory)	Parents find the reports useful and children don't notice the chirps	Parent interviews; audibility check with parental permission	No pass/fail; report findings

The 63 mimic subjects are CAP's periodic leg movement (10), sleep-disordered breathing (4), nocturnal frontal lobe epilepsy (40) and insomnia (9) groups.

System overview

One sensing engine turns echoes into event streams; a shared feature layer and sleep stager feed three modules, and everything runs on the phone.

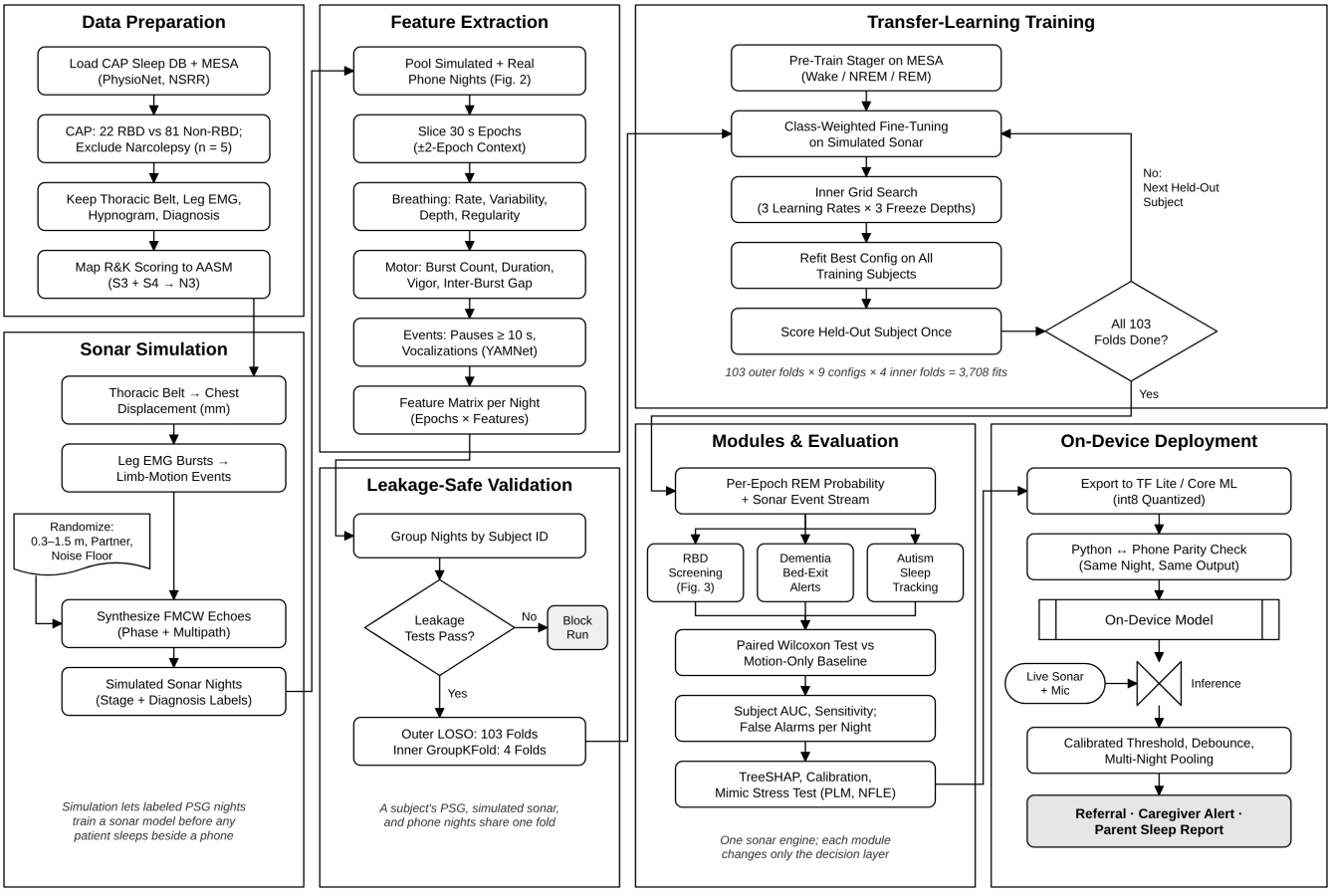


Figure 1. NightEcho pipeline, from public sleep-lab data and phone recordings to on-device alerts.

1. **Data preparation.** Public sleep-lab recordings with confirmed diagnoses (CAP Sleep Database) plus a large sleep-study set for pre-training (MESA, or the open PhysioNet 2018 set if MESA access is slow).
2. **Sonar simulation.** Breathing-belt and leg-muscle signals become sonar-like signals, keeping the sleep-stage and diagnosis labels.
3. **Feature extraction.** Each 30-second epoch gets breathing, movement, event and time-of-night features.
4. **Leakage-safe validation.** A subject's data never appears in both training and testing.
5. **Transfer-learning training.** A sleep stager learns on real sleep-study breathing, then adapts to sonar.
6. **Modules and evaluation.** RBD screening, dementia bed-exit alerts and autism sleep tracking, each tested against a simpler baseline.
7. **On-device deployment.** The trained models run on the phone, and only results leave the app.

Two tracks run in parallel: the **data track** (Phases 1–7, on laptops) and the **phone track** (Phase 8). They meet when the trained model is exported into the app.

Sonar sensing design

The phone sweeps an 18–22 kHz chirp every 10 ms. Echoes are sorted into 4.3 cm range bins, and tiny phase shifts in the sleeper's bin reveal each breath.

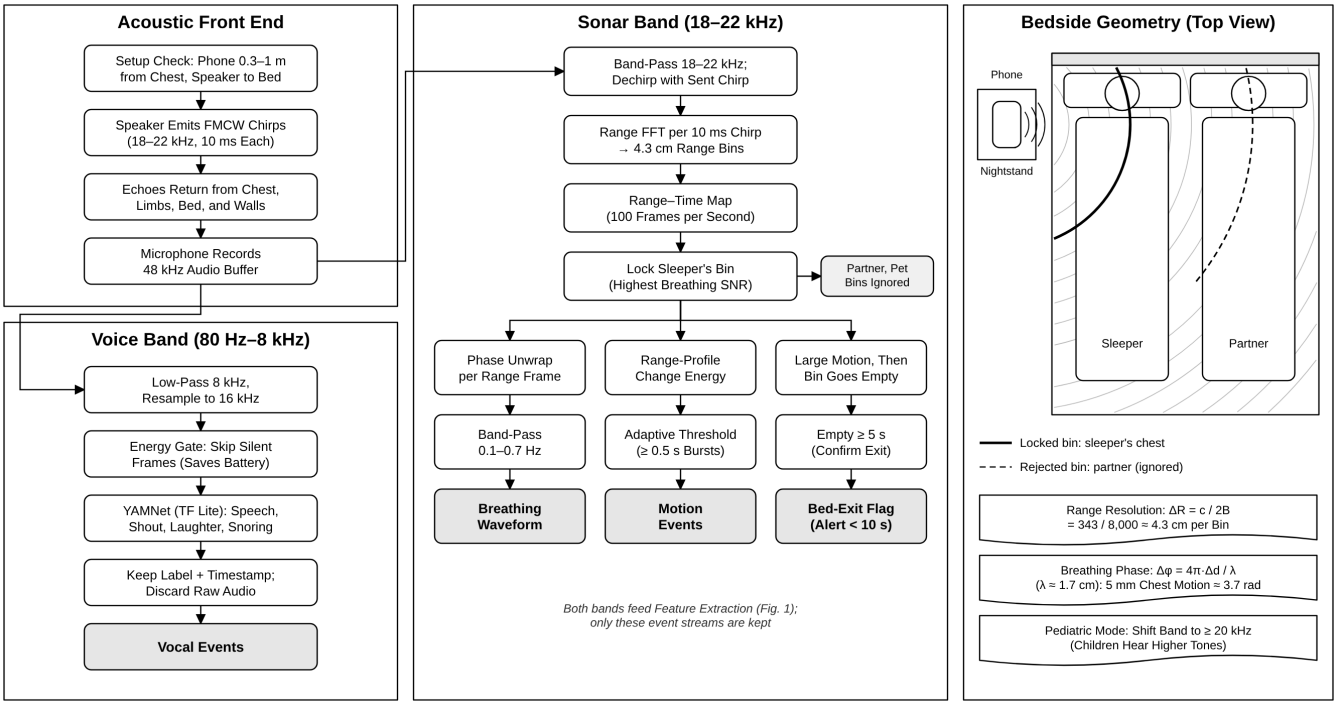


Figure 2. Smartphone sonar front end: acoustic chain, sonar and voice bands, and bedside geometry.

Parameters

Parameter	Value	Why
Sweep band	18–22 kHz (pediatric mode: 20 kHz and up)	Above most adults' hearing; inside a phone's 48 kHz capture
Chirp length	10 ms (480 samples), edges tapered over 0.5 ms	100 frames per second; tapering prevents audible clicks
Sample rate	48 kHz, mono, 16-bit	Supported by nearly all phones; captures up to 24 kHz
Range bin	≈ 4.3 cm	Set by the 4 kHz sweep bandwidth
Range window	0.2–1.5 m	Bedside distances; an echo from 1.5 m returns in 8.7 ms, inside one chirp
Wavelength	≈ 1.7 cm at 20 kHz	5 mm of chest motion shifts the echo phase by ≈ 3.7 rad
Breathing band	0.1–0.7 Hz	6–42 breaths per minute, adults through young children
Motion rule	Change energy above median + 5 × MAD for ⩾ 0.5 s	Adapts to each bedroom's noise level
Bed-exit rule	Sleeper's bin empty ⩾ 5 s after large motion	Leaves time to alert in under 10 s
Voice band	80 Hz–8 kHz, resampled to 16 kHz	The input format of YAMNet, the sound classifier

Key equations

Range resolution, where c is the speed of sound and B the sweep bandwidth:

```
\Delta R = \frac{c}{2B} = \frac{343\ \mathrm{m/s}}{2 \times 4000\
\mathrm{Hz}} \approx 4.3\ \mathrm{cm}
```

Breathing phase, where Δd is chest displacement and λ the wavelength:

$$\Delta\varphi = \frac{4\pi\,\Delta d}{\lambda}, \qquad \lambda = \frac{c}{f} = \frac{343}{20\,000} \approx 1.7\ \mathrm{cm}, \qquad \Delta d = 5\ \mathrm{mm} \Rightarrow \Delta\varphi \approx 3.7\ \mathrm{rad}$$

The signal chain, in order

1. **Generate the chirp:** a linear sweep from 18 to 22 kHz over 10 ms, with a 0.5 ms raised-cosine taper at both ends, looped continuously at low volume.
2. **Align frames:** cross-correlate the recording with the chirp. The first strong peak is the direct speaker-to-microphone path, which marks where each frame starts.
3. **Dechirp:** band-pass 18–22 kHz, form the analytic (complex) signal, and multiply by the complex conjugate of the transmitted chirp.
4. **Range FFT:** one FFT per 10 ms frame gives a complex range profile with 4.3 cm bins.
5. **Remove clutter:** subtract each bin's 10-second moving average, which erases walls, the bed frame and other still objects.
6. **Lock the sleeper's bin:** between 0.2 and 1.5 m, pick the bin with the most energy in the 0.1–0.7 Hz breathing band over 30 s. Switch only if another bin stays stronger for 60 s.
7. **Breathing:** unwrap the phase of the locked bin, band-pass it to 0.1–0.7 Hz and convert it to displacement ($\Delta d = \lambda \Delta\varphi / 4\pi$).
8. **Movement:** sum the frame-to-frame change across bins from 0.2 to 1.5 m, and flag bursts above median + 5 × MAD lasting at least 0.5 s.
9. **Bed exit:** large movement, then the sleeper's bin stays empty for at least 5 s, triggers an alert.
10. **Voice:** low-pass to 8 kHz, resample to 16 kHz and run YAMNet. Keep only the labels and timestamps, then discard the audio.

Known limits and safeguards

- **Obstructive apnea:** chest effort can continue while airflow stops, so sonar sees falling breathing amplitude and a large recovery breath, not a flat line. The breathing-pause check therefore looks for amplitude falling at least 50% for 10 s or more, followed by a recovery breath.
- **Children:** young people often hear tones near 20 kHz. Pediatric mode sweeps 20–22 kHz, which coarsens the range bins to 8.6 cm, and an audibility check runs before the first night.

- **Pets:** dogs and cats hear well above 22 kHz. Keep the volume low and watch for reactions during testing.
- **Phone differences:** speakers and microphones fade near 20 kHz at different rates. Measure each test phone first (step 8.2); if a phone can't reach 22 kHz, sweep 18–20 kHz instead.
- **Power and heat:** overnight playback and processing need the phone plugged in. Log battery temperature during the overnight test.
- **REM in people with RBD:** movement during REM can make REM look like wake to a sleep stager (Skjærbæk et al., 2026), so the stager leans on breathing irregularity and sleep-cycle timing (Phase 4).

Data sources and access

The two open PhysioNet datasets can be downloaded today. MESA needs a free online request, so submit it in week 1 and use the PhysioNet 2018 set until it's approved.

Dataset	What we use	Size	Access	First step
CAP Sleep Database (PhysioNet)	108 sleep studies: 22 RBD, 16 healthy, 70 other sleep disorders. Chest and abdominal belts, airflow, leg EMG on both legs, R&K sleep stages	≈ 40.1 GB	Open (Open Data Commons Attribution 1.0); cite Terzano et al., 2001	Download with the `wfdb` Python package or `wget`
MESA Sleep (National Sleep Research Resource)	2,237 participants in the sleep exam: overnight sleep study plus 7 days of wrist actigraphy	Large; download only the channels you need	Free; online Data Access and Use Agreement; approval time varies	Create a sleepdata.org account and submit the request in week 1

Dataset	What we use	Size	Access	First step
PhysioNet 2018 Challenge	994 training subjects with 30-second sleep stages; 13 channels at 200 Hz, including chest, abdomen and airflow	≈ 135 GB (training set)	Open (Open Data Commons Attribution 1.0)	Backup for MESA; start with 200 nights
YAMNet (pre-trained sound model)	Detects speech, shouting, laughter and snoring	Small	Free (TensorFlow)	Use the TensorFlow Lite version on the phone
Team-collected nights	Phone sonar alongside a Vernier respiration belt; scripted trials with video as ground truth	Small	Signed consent (see Ethics)	Start after consent forms are signed
Clinical pilot (future)	Phone recording beside a video sleep study	—	Partner institution's IRB	Through a clinical partner (Phase 10)

Data-handling rules:

- Keep raw datasets on an external drive. Never commit them to GitHub; they are too large, and data use agreements typically forbid redistribution.
- Never try to identify people in any dataset.
- Cite every dataset in the brief, the video credits and the website.
- Record every excluded night and the reason in the data card (step 1.8).

Step-by-step build: Phases 0–3

Phases 0–3 build the data foundation: by December 6 every public sleep study is cleaned, converted into simulated sonar, and turned into features.

Phase 0 · Setup and requests (Nov 2–8)

- ☐ **0.1** Open the Award Force portal and confirm your Lean Canvas was submitted and accepted. If it wasn't submitted by October 29, email info@conradchallenge.org today.
- ☐ **0.2** Assign the four roles (see Team roles). Set a weekly team meeting and a check-in with your coach every two weeks.
- ☐ **0.3** Create a private GitHub repository with folders `data/`, `sim/`, `features/`, `models/`, `app/` and `docs/`, a README, and a dated lab notebook (`docs/notebook.md`) with one entry per work session.
- ☐ **0.4** Set up Python 3.11 or later in a virtual environment with numpy, scipy, pandas, wfdb, pyedflib, scikit-learn, xgboost, torch, shap and matplotlib.
- ☐ **0.5** Create a sleepdata.org account and submit the MESA data request. Describe the use plainly: training a contactless sleep-stage model for a student research project.
- ☐ **0.6** Run the novelty check (Prior work section) and start the prior-art log.
- ☐ **0.7** Email 5–10 sleep neurologists or sleep-lab directors at nearby universities and hospitals to ask for mentorship (see Phase 10).
- ☐ **0.8** Order the Vernier Go Direct Respiration Belt and one used Android test phone (see Budget).
- ☐ **0.9** Get a 1 TB external drive for the datasets.

Done when a test script opens one sleep-study file, the MESA request is submitted, mentor emails are sent and equipment is ordered.

Phase 1 · Public-data pipeline (Nov 2–22)

- ☐ **1.1** Download CAP to the external drive with the `wfdb` package or `wget`.
- ☐ **1.2** Write `load_night(record)`: read the EDF file, list its channels, and map them to standard names (`thorax`, `abdomen`, `airflow`, `tib_left`, `tib_right`). Log any record that is missing a channel.
- ☐ **1.3** Parse the sleep-stage files into 30-second labels and map the old R&K stages to today's AASM stages: S1 → N1, S2 → N2, S3 and S4 → N3, REM → REM, movement time → Wake. Drop unscored epochs.

- [ ] **1.4** Build `subjects.csv` with record ID, diagnosis, age, sex (from CAP's age-sex spreadsheet), channels present and include/exclude. Exclude the 5 narcolepsy recordings, because narcolepsy often comes with RBD-like behavior and would blur the labels.
- [ ] **1.5** Quality check: plot one night per diagnosis (chest belt, both legs, sleep stages) and note flat or noisy channels.
- [ ] **1.6** Preprocess: low-pass the chest belt at 2 Hz and resample to 10 Hz. High-pass leg EMG at 10 Hz, rectify it, smooth it into an envelope and resample to 10 Hz. Save each night as one `.npz` file with its labels.
- [ ] **1.7** Repeat 1.2–1.6 for the pre-training set: MESA if approved, otherwise 200 nights from PhysioNet 2018 (chest belt and sleep stages).
- [ ] **1.8** Write a one-page data card: counts per group, exclusions and reasons, and the share of epochs in each sleep stage.

Done when every included night loads with one function call and the data card is complete.

Phase 2 · PSG-to-sonar simulator (Nov 9–Dec 6)

- [ ] **2.1** Chest motion: z-score each night's chest belt and scale it to displacement with a random breathing depth of 3–10 mm.
- [ ] **2.2** Limb movements: detect leg-EMG events with the AASM limb-movement rule (at least 8 μV above resting EMG, lasting 0.5–10 s).
- [ ] **2.3** Turn each limb movement into a moving reflector with a random speed of 0.2–1 m/s, placed 0.3–0.8 m beyond the chest, plus a small disturbance in the chest bin.
- [ ] **2.4** Build the range–time map directly at 100 frames per second instead of simulating raw audio, which is far faster (equation below).
- [ ] **2.5** Randomize each night: phone-to-chest distance 0.3–1.5 m; a bed partner half the time (breathing borrowed from another night, 0.3–0.8 m farther away); occasional pet movement; noise level; and a gentle gain tilt across bins to mimic different phones.
- [ ] **2.6** Make 3 randomized versions of each night. Every version keeps its subject's ID.
- [ ] **2.7** Sanity check: run the Phase 3 signal chain on simulated maps. Recovered breathing should match the belt ($r \geq 0.9$), and at least 90% of injected limb movements should be found.
- [ ] **2.8** Save the simulated nights with their inherited labels: sleep stage per epoch, diagnosis and event times.

For each reflector i at range R_i in frame n, with K spreading its energy across neighboring bins, C the static clutter and ε the noise:

```
X[n,k] = \sum_i a_i\, K\big(k - R_i(n)/\Delta R\big)\, e^{\,j 4\pi
R_i(n)/\lambda} + C[k] + \varepsilon[n,k]
```

Done when step 2.7 passes on 20 random nights drawn from every diagnosis group.

Phase 3 · Feature extraction (Nov 23–Dec 6)

- ☐ **3.1** Implement the shared chain from the range–time map onward (Sonar sensing design, steps 5–9) as one Python module. The phone app will copy it exactly.
- ☐ **3.2** Breathing features per 30-second epoch: rate (spectral peak), breath-to-breath variability, depth (median peak-to-peak), regularity (autocorrelation peak) and change in depth from the previous two epochs.
- ☐ **3.3** Movement features: burst count, total duration, peak energy (vigor), time since the last burst, and whether bursts repeat every 5–90 s (the periodic leg movement pattern).
- ☐ **3.4** Breathing-disturbance features: drops in amplitude of at least 50% lasting 10 s or more, and the recovery breaths that follow.
- ☐ **3.5** Vocal features (real nights only): YAMNet counts per epoch for speech, shouting, laughter, crying and snoring.
- ☐ **3.6** Context features: minutes since sleep onset, clock hour, and the two epochs before and after.
- ☐ **3.7** Unit-test each feature on synthetic signals with known answers; for example, a 0.25 Hz sine wave must give 15 breaths per minute.
- ☐ **3.8** Save one feature table per night (Parquet format).

Done when all unit tests pass and features exist for every simulated night.

Step-by-step build: Phases 4–6

Phases 4–6 turn features into decisions: a sleep stager that finds REM-like sleep, the RBD screening logic, and the dementia and autism modules.

Phase 4 · Sleep staging (Dec 7–20)

- ☐ **4.1** Build a baseline first: gradient-boosted trees on the epoch features, predicting Wake, NREM or REM. This guarantees a working result early.

- [ ] **4.2** Build the main model: a 1-D convolutional network reads each epoch's 10 Hz breathing and movement signals, and a bidirectional LSTM reads the sequence of epochs across the night. Output: Wake, NREM or REM.
- [ ] **4.3** Use a class-weighted loss, because REM is the minority stage.
- [ ] **4.4** Pre-train on MESA (or PhysioNet 2018) chest-belt signals passed through the simulator.
- [ ] **4.5** Fine-tune on CAP simulated sonar, using only the training subjects inside each leave-one-subject-out fold. Inner grid: 3 learning rates × 3 choices of how many layers to freeze.
- [ ] **4.6** Check the RBD trap: movement during REM makes REM look like wake. Measure REM recall separately in RBD subjects. If it's low, add features for breathing irregularity and time since the last REM-like period, since sleep cycles repeat roughly every 90 minutes.
- [ ] **4.7** Output a calibrated REM probability per epoch (temperature scaling, fitted on inner folds).
- [ ] **4.8** Report Cohen's kappa against the sleep-lab stages, recall per stage and a confusion matrix, separately for the RBD, mimic and healthy groups.

Done when every held-out subject has a REM probability for every epoch.

Phase 5 · RBD screening logic (Dec 14–27)

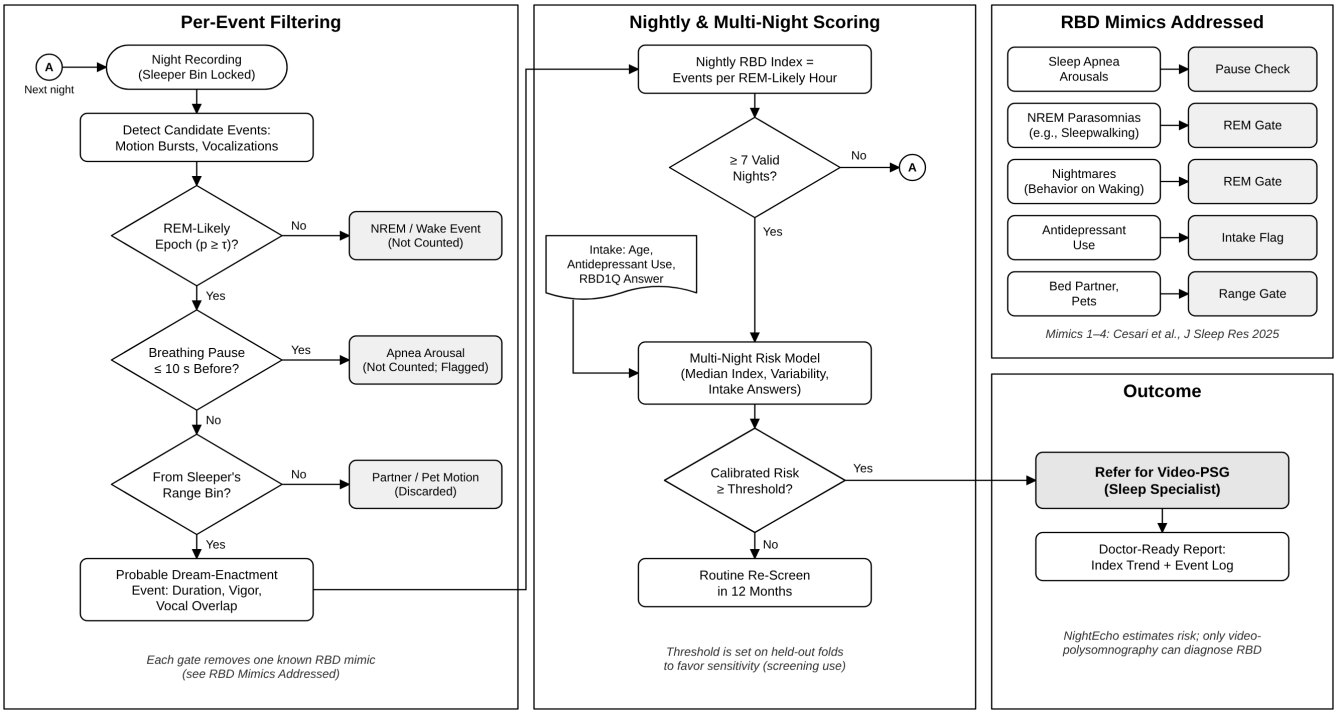


Figure 3. RBD screening logic: per-event gates, nightly index and multi-night decision.

- [ ] **5.1** Collect candidate events: movement bursts (all nights) and vocal events (real nights).
- [ ] **5.2** REM gate: keep events in epochs whose REM probability is at least τ. Choose τ on inner folds only.
- [ ] **5.3** Breathing-pause check: drop events that start within 10 s after a breathing disturbance ends, and log them for the breathing summary.
- [ ] **5.4** Range gate: keep events from the sleeper's bins and leg zone; drop partner and pet bins.
- [ ] **5.5** Nightly RBD index = surviving events per hour of REM-like sleep. Add night features: event duration and vigor, overlap with vocal events, share of events in the second half of the night, and the rate and periodicity of NREM movements (which separates periodic leg movements).
- [ ] **5.6** Define a valid night: at least 4 hours of detected sleep, with the sleeper's bin locked for at least 80% of it.
- [ ] **5.7** Subject-level inputs: the median and spread of the nightly index across valid nights, plus intake answers (age, antidepressant use, the RBD single-question screen). CAP provides only age, so the other answers enter in the clinical pilot.
- [ ] **5.8** Train two classifiers: L2-regularized logistic regression (baseline) and XGBoost (main), both class-weighted. Calibrate with Platt scaling on inner folds.
- [ ] **5.9** Pick the threshold on inner folds to reach at least 90% sensitivity. In screening, a missed case costs more than a referral that rules RBD out.
- [ ] **5.10** State CAP's limit plainly: it has one night per person, so Stage 1 tests the single-night index. The 3 simulated versions of each night test robustness to setup, not real night-to-night variation, which needs the clinical pilot.

Done when all 103 held-out subjects have a calibrated risk and a refer or re-screen decision.

Phase 6 · Dementia and autism modules (Dec 14–Jan 3)

Dementia module:

- [ ] **6.1** Bed-exit detector: large movement, then the sleeper's bin empty for at least 5 s, triggers an alert. Cancel it if breathing returns within those 5 s.
- [ ] **6.2** Send alerts to the caregiver's phone with Firebase Cloud Messaging, and measure the end-to-end delay.

- [ ] **6.3** Log time out of bed and the return, and show a morning summary.
- [ ] **6.4** Build a weekly trend for clinicians: wake time after sleep onset, number of awakenings and sleep efficiency. Keep it descriptive, with no diagnosis.
- [ ] **6.5** Add caregiver settings: quiet hours, alert delay, and a second caregiver if the first doesn't respond.

Autism module:

- [ ] **6.6** Pediatric mode: sweep 20–22 kHz at lower volume, and run an audibility check before the first night (play the sweep while the child is awake; the parent watches for any reaction).
- [ ] **6.7** Measure sleep-onset latency (from a "lights out" tap to the first 10 minutes of steady sleep), night wakings and total sleep time.
- [ ] **6.8** Build a weekly parent report with tags for routine changes (melatonin, bedtime routine, illness) to share with clinicians. Make no diagnostic claims.
- [ ] **6.9** Interview at least 5 parents of autistic children and 1–2 occupational therapists about which report views help, and record what you changed.

Done when both modules run end to end on simulated nights and scripted-trial recordings.

Step-by-step build: Phases 7–8

Phase 7 makes the results trustworthy, and Phase 8 puts the engine on a real phone. They run in parallel with the modeling work.

Phase 7 · Validation statistics (Dec 7–Jan 3)

- [ ] **7.1** Lock the splits before training anything: the subject list, 103 outer folds (one per subject) and 4 inner folds grouped by subject. Save them as `splits.json` and never change them after seeing results.
- [ ] **7.2** Write automated leakage tests that stop a run if a subject appears in both training and testing, if a subject's simulated versions are split across folds, or if any scaling, threshold or setting was fit on test data.
- [ ] **7.3** Pre-register the analysis: before the final runs, commit a dated `analysis_plan.md` naming the primary outcome, metrics, tests and threshold rule. Posting it on OSF (Open Science Framework) adds a public timestamp.

- [ ] **7.4** Primary metric: subject-level AUC with a 95% bootstrap confidence interval (2,000 resamples of subjects).
- [ ] **7.5** Secondary metrics: sensitivity and specificity at the chosen threshold; the false-positive rate in each mimic group; calibration (Brier score and a reliability plot); and staging kappa.
- [ ] **7.6** Comparisons: the full model against a motion-only baseline, using the DeLong test for the AUC difference and a paired Wilcoxon signed-rank test on per-subject Brier scores. Then the model with and without gates, using McNemar's test on false positives.
- [ ] **7.7** Explain the model with TreeSHAP and check the top features make clinical sense. For example, events inside REM windows should matter more than NREM events.
- [ ] **7.8** Make the results figure and table for the brief, and write one honest paragraph on the study's limits.

Compute: 103 outer folds × 9 settings × 4 inner folds = 3,708 small model fits. Free GPU notebooks (Kaggle, Google Colab) can handle this; run the baselines on a laptop CPU.

Done when every metric has a confidence interval and both comparisons are finished.

Phase 8 · Phone app (Nov 16–Jan 3)

Stage A, no app needed (first week):

- [ ] **8.1** In Python, generate a 10-minute 48 kHz WAV file of looping chirps (18–22 kHz, 10 ms each, 0.5 ms tapers).
- [ ] **8.2** On each test phone, play the WAV at low volume while a recorder app saves uncompressed 48 kHz audio. Check the spectrogram: is the 18–22 kHz sweep there, and is the direct speaker-to-microphone path visible? Choose the band for each phone.
- [ ] **8.3** Record 10 minutes of a teammate lying still and breathing normally, and process it with the Phase 3 code. Breathing in the phase plot is your first proof of concept; film it for the video.

Stage B, native Android prototype (Kotlin):

- [ ] **8.4** Play the chirp with `AudioTrack` and record with `AudioRecord` at 48 kHz, using the `UNPROCESSED` audio source if the phone supports it (otherwise `VOICE_RECOGNITION`). Run both in a foreground service so they continue all night with the screen off.
- [ ] **8.5** Align frames to the direct speaker-to-microphone path with cross-correlation.

- [ ] **8.6** Port the signal chain (dechirp, FFT, clutter removal, bin lock, phase, motion energy) to Kotlin or C++. Run the parity check: the same recording processed in Python and on the phone must give matching outputs.
- [ ] **8.7** Export the stager and RBD model to TensorFlow Lite (int8) and run them every 30 s. Run YAMNet's TensorFlow Lite version for vocal events, keeping only labels and timestamps.
- [ ] **8.8** Store nightly features in an encrypted on-phone database. Upload nothing unless the user opts in.
- [ ] **8.9** Build the screens: setup check (distance and volume), start and stop a night, morning summary, weekly trends, caregiver alerts, pediatric-mode switch and intake questions (age, medications, the RBD single-question screen).
- [ ] **8.10** Run an 8-hour overnight test with the phone plugged in, logging dropped frames and battery temperature.
- [ ] **8.11** Optional, once Android works: an iOS version using `AVAudioEngine`, with the audio session in measurement mode to minimize the system's own audio processing.

Done when an 8-hour night completes with under 1% dropped frames and the parity check passes.

Step-by-step build: Phases 9–11

Phases 9–11 test NightEcho on people, line up a clinical partner and package everything for Conrad's judges.

Phase 9 · Human testing (Dec 14–Feb 28)

- [ ] **9.1** Write consent forms: adult consent, and for anyone under 18, written parental permission plus the minor's own assent. If you also enter an ISEF-affiliated science fair, an IRB must approve the study before you recruit anyone (ISEF rules).
- [ ] **9.2** Write a data plan: what you record, where it's stored (encrypted), who can see it and when it's deleted. Delete trial videos once they're annotated.
- [ ] **9.3** Stage 2 home nights: 10 healthy adults × 2 nights, phone on the nightstand plus the Vernier belt. Sync the clocks at the start with three deep breaths visible in both recordings, and include bed partners when possible.
- [ ] **9.4** Analyze Stage 2: breathing rate per 30-second epoch, phone vs belt. Make a Bland–Altman plot (bias and 95% limits of agreement). Target: mean absolute error under 1 breath per minute.

- [ ] **9.5** Stage 3 scripted trials: per session, 20 acted dream-enactment movements (punching, kicking, talking, shouting) and 20 bed exits, with and without a partner. A separate camera records video as ground truth.
- [ ] **9.6** Analyze Stage 3: event recall, false alarms per hour of quiet lying, and alert delay (video exit time to alert timestamp).
- [ ] **9.7** Pediatric audibility check, with parental permission and the child's assent: play the sweep while the child is awake, and adjust band and volume until there's no reaction.
- [ ] **9.8** Note any pet reactions during home nights, and lower the volume if needed.

Done when the Stage 2 and Stage 3 targets are evaluated and logged, pass or fail.

Phase 10 · Clinical partner (Nov 2–Apr 20)

- [ ] **10.1** Shortlist 10 sleep centers or movement-disorder clinics, starting with university hospitals. Look for clinicians who publish on RBD.
- [ ] **10.2** Send a short email: who you are, the one-sentence idea, Figure 1, a request for a 20-minute call, and a promise to follow their institution's rules. Follow up once after a week.
- [ ] **10.3** Ask for three things, in order: mentorship, a letter of support for the brief, and a pilot.
- [ ] **10.4** Draft the Stage 4 pilot protocol with the partner: adults with RBD confirmed by video sleep study, plus controls, with the phone beside the bed during their scheduled study. Outcomes are AUC, specificity and calibration. With 30 RBD patients and 30 controls, an AUC near 0.85 has a 95% confidence interval of about ±0.10 (Hanley–McNeil method).
- [ ] **10.5** The partner's IRB approves the pilot, and the partner handles patient data under its own privacy rules.
- [ ] **10.6** Attach letters of support to the Innovation Brief as optional attachments.

Done when at least one clinician agrees to mentor you or write a letter.

Phase 11 · Conrad deliverables (Dec 14–Jan 7, then Summit prep)

- [ ] **11.1** Innovation Brief (at most 3,000 words across 10 questions; see the mapping section): full draft Dec 20, coach review Dec 27, final Jan 3.
- [ ] **11.2** Attach the reference list, plus up to two optional attachments (Figures 1 and 3, or a one-page results sheet and letters).

- [ ] **11.3** Innovation Video (3–5 minutes): storyboard by Dec 15. Film the app on a real nightstand, the breathing proof from step 8.3, the figures, results and impact. Finish editing by Dec 30, then upload an MP4 under 800 MB and an unlisted YouTube or Vimeo link.
- [ ] **11.4** Website: team story, mission, figures and brand (name, logo, colors, voice), reachable by direct link (you may hide it from search engines). Live by Jan 3.
- [ ] **11.5** Pay the $499 entry fee or apply for financial aid. Submit everything in Award Force by Jan 5, two days before the Jan 7, 11:59 PM CT deadline, and test the video and website links from a logged-out browser.
- [ ] **11.6** If you're named finalists in February: prepare the Power Pitch and executive brief, build the tabletop display, and rehearse answers to 20 hard judge questions, such as "Isn't phone sonar already done?", "How do you know it's REM?" and "What about privacy and the FDA?"
- [ ] **11.7** Innovation Summit: Apr 21–24, 2027, Space Center Houston.

Done when everything is submitted by January 5.

Validation plan and targets

Validation climbs four stages, from a simulation the team can run now to a clinical pilot that needs a partner. Each stage must meet its target before the next begins.

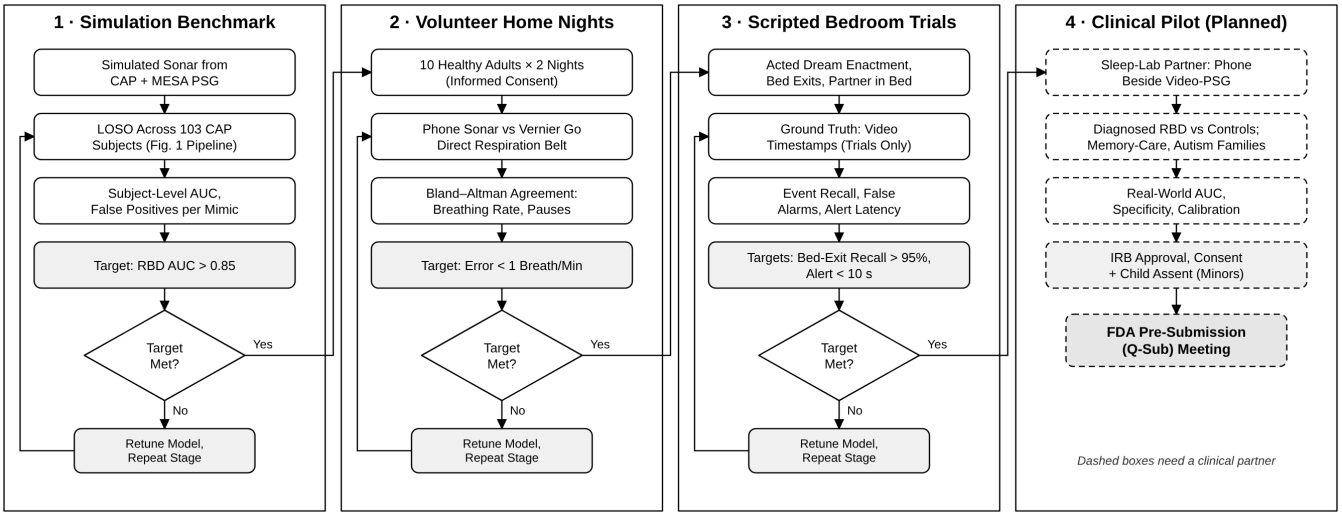


Figure 4. Staged validation plan with pass/fail gates.

Stage	Data	Primary metric	Target	What it needs
1 · Simulation	103 CAP subjects as simulated sonar	Subject-level AUC, leave-one-subject-out	> 0.85	Laptops and free GPU time
2 · Home nights	10 adults × 2 nights, phone plus belt	Breathing-rate error	< 1 breath/min	Consent forms; respiration belt
3 · Scripted trials	Acted movements and bed exits, partner present	Bed-exit recall; alert delay	> 95%; < 10 s	Consent; a camera for ground truth
4 · Clinical pilot	RBD confirmed by video sleep study, vs controls	Real-world AUC, specificity, calibration	Set with the partner	Clinical partner and IRB approval

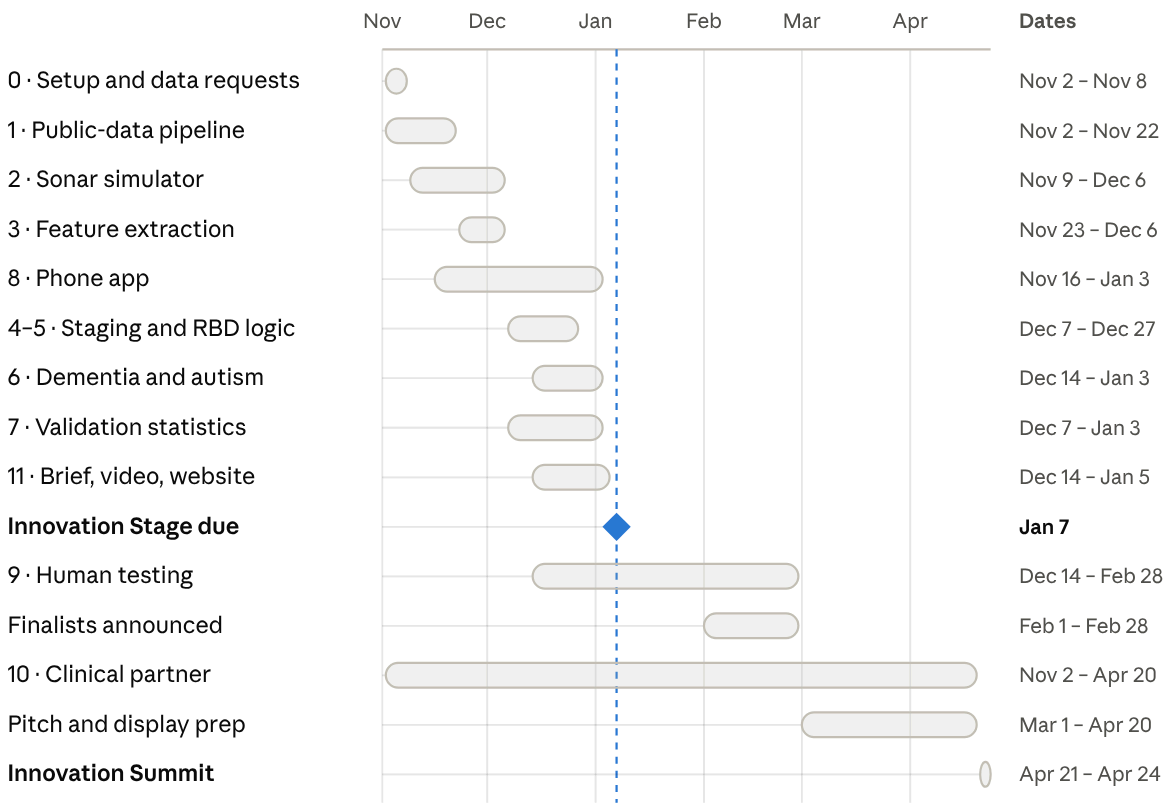
By January 7, aim to have Stage 1 complete, several Stage 2 nights analyzed, one Stage 3 session done, and Stage 4 written up as a plan with your outreach status.

If a target is missed, report the result with its confidence interval, the likely reason and the fix you'll try next. An honest miss with a clear next step is more credible than a vague claim.

Timeline

Everything judged is due January 7; testing runs to the April Summit

NightEcho build plan by phase, November 2026 to April 2027



NightEcho build plan · November 2026 to April 2027

The data track (Phases 0–7) and the phone track (Phase 8) run in parallel through December, so a working demo and Stage 1 results exist before the January 7 submission. Human testing and clinical outreach continue toward the Summit. Dates assume work starts Monday, November 2; the exact February announcement date isn't published, so it's shown as the whole month.

Team roles

Conrad teams have 2–5 students plus an adult coach. Four roles cover the work; a smaller team combines them.

Role	Owns	Phases
Signal-processing lead	Sonar chain, simulator, phone-response tests	2, 3, 8 (with the app lead)
Machine-learning lead	Sleep stager, RBD model, leakage tests, statistics	4, 5, 7
App lead	Android app, on-device models, caregiver alerts, parent reports	6, 8
Research and business lead	Literature, novelty check, ethics forms, testing logistics, clinical outreach, brief, video, website	0, 9, 10, 11
Coach (adult)	Weekly guidance, consent and data-plan review, budget, Conrad portal	All

- **Two or three students:** combine signal processing with the app, and machine learning with research and business.
- **Five students:** add a design lead for the video, website, logo and Summit display.
- **Weekly rhythm:** a 15-minute planning call on Monday, work sessions midweek, and a Sunday demo plus lab-notebook entries.

Budget

The prototype costs about $433 with every optional item, which keeps it under the $500 in the Lean Canvas. The largest single cost is Conrad's $499 entry fee, and financial aid is available for it.

Item	Cost (USD)	Notes
Vernier Go Direct Respiration Belt	$149	Breathing reference for Stage 2 (vernier.com)
Used Android test phone	about $100 (estimate)	A second brand to compare speakers; team phones cover the rest
External 1 TB drive	about $60 (estimate)	Holds the datasets

Item	Cost (USD)	Notes
Google Play developer account	$25, one time	Optional: needed only to distribute the app, not to test it
Apple Developer Program	$99 per year	Optional: only for the iOS version
Cloud computing	$0	Free GPU notebooks (Kaggle, Google Colab)
Prototype subtotal	**about $433**	With every optional item
Innovation Stage entry fee	$499	Due with the January 7 submission; financial aid available
Website domain	about $15 per year (estimate)	Optional; free site builders work
Summit display printing	about $100 (estimate)	Only if named finalists
Travel to Houston	Varies	Only if named finalists

Possible funding: school club or parent-organization grants, local Parkinson's or Alzheimer's community groups, Conrad financial aid, and crowdfunding. Check current prices before buying.

Ethics, privacy, safety and regulation

NightEcho handles sensitive health information about vulnerable people, so these rules shape the design itself, not just the paperwork.

People in the study

- Adult volunteers sign informed consent. Minors need written parental permission plus their own assent.
- If you enter an ISEF-affiliated fair, an IRB must approve human testing before you recruit. Studies that only review publicly available datasets such as CAP are exempt under ISEF rules; ask your fair's review committee whether a request-access dataset like MESA counts.
- NightEcho never tells anyone they have a disease. Results say "higher risk" and recommend a doctor visit.

- Hearing "higher risk of Parkinson's" is frightening, and there is no proven prevention yet. Explain what RBD risk means in plain words, avoid alarming language, and in the clinical pilot let a clinician deliver results.

Privacy and data security

- Raw audio is never saved or uploaded. Only event labels, timestamps and nightly features are kept.
- Processing happens on the phone, local storage is encrypted, and any upload is opt-in.
- Consumer health apps outside HIPAA fall under the FTC's Health Breach Notification Rule, amended in 2024, which counts sharing health data without permission as a breach.
- If the free tier shows ads, use contextual ads only: never use health data for targeting, and never include an ad tool that collects it. The FTC has penalized health apps for sharing health data with advertisers.
- Patient data from a clinical partner stays under that clinic's HIPAA rules and data agreements.

Safety

- Keep the chirp volume low. Use pediatric mode for children, who hear higher tones than adults, and watch pets, which hear well above 22 kHz.
- Present bed-exit alerts as a convenience, not a medical alarm, until testing proves them reliable.
- Keep the phone plugged in on the nightstand, never under a pillow, to avoid overheating.

Regulation

- An app that screens for disease risk is likely regulated by the FDA as medical-device software, while basic sleep tracking faces lighter rules. Until cleared, present NightEcho as a research prototype.
- The likely path: an FDA Pre-Submission (Q-Sub) meeting backed by clinical-pilot data, then a De Novo or 510(k) request.

Business plan summary

Families can use NightEcho free or by subscription; the revenue that scales comes from memory-care facilities, drug-trial sponsors and licensing. All prices are hypotheses to test.

Market

Segment	Size	Who pays
Adults with RBD	1.06% of adults in a Swiss general-population study, mostly undiagnosed	Screening is free; Parkinson's trial sponsors pay per referral
People with Alzheimer's dementia	7.4 million Americans age 65+ in 2026	Families (subscription); memory-care facilities (per bed)
Autistic children	About 1 in 31 U.S. eight-year-olds	Families (subscription)

Revenue streams (from the Lean Canvas)

- Free screening tier supported by contextual ads, with no health data shared.
- $7.99 per month premium: no ads, caregiver alerts and doctor-ready reports.
- Memory-care facilities: $15 per bed per month, running on refurbished phones.
- Per-referral fees from Parkinson's trial sponsors.
- Licensing the sonar engine to sleep apps.

Because processing runs on the phone, the cost of serving one more user is low; the main costs are clinical validation, regulatory work, security and support.

Go-to-market steps

1. Run a free pilot with 5–10 families from local autism and dementia caregiver support groups.
2. Partner with one sleep clinic for the RBD pilot and a referral pathway.
3. Pilot in one memory-care facility using refurbished phones.
4. Approach Parkinson's research groups once the clinical pilot has data.

5. Grow through sleep clinics, neurologists, Parkinson's and Alzheimer's foundations, autism parent networks and the app stores.

Competitors and positioning

Competitor	What it offers	NightEcho's difference
SleepScore app	Phone-sonar sleep tracking	RBD screening, dementia alerts and a pediatric mode
Under-mattress sensors (e.g., Withings)	Bed-sensor sleep tracking	Nothing to buy or install
Bedroom radar (e.g., MinderCare research)	Contactless dementia monitoring	No added hardware; RBD screening
Smartwatches and rings	Wrist or finger sleep tracking	Nothing to wear or charge, which matters for children and people with dementia
Sleep labs	Diagnosis	NightEcho decides who needs a lab study, sending them more of the right patients

Risks and mitigations

The biggest risks are scientific: simulated sonar may not match real sonar, and REM is hard to detect in people with RBD. Both are tested early and reported honestly.

Risk	Likelihood (our estimate)	Impact	Mitigation
Simulated sonar doesn't match real phone sonar	Medium	High	Tune the simulator with real phone recordings (step 8.3); label Stage 1 results as simulation; confirm on real nights in Stages 2–4

Risk	Likelihood (our estimate)	Impact	Mitigation
REM detection is weak, especially in RBD	High	High	Lean on breathing irregularity and sleep-cycle timing; report REM recall per group; fall back to a whole-night movement index if needed
Only 22 RBD subjects in CAP	High	Medium	Confidence intervals, simple models, a pre-registered plan, and framing Stage 1 as feasibility
Time crunch before January 7	High	High	Build baselines first; submit by January 5; cut iOS and extras first
No clinical partner by January	High	Medium	Present Stage 4 as a plan with an outreach log; keep asking through April
A phone can't play or record near 22 kHz	Medium	Medium	Test each phone first (step 8.2); narrow the sweep to 18–20 kHz
Children or pets hear the chirps	Medium	Medium	Pediatric mode, low volume and audibility checks
The phone stops the app overnight to save battery	Medium	Medium	Foreground service, phone plugged in, battery optimization turned off for the app; overnight test
Results frighten users	Medium	High	"Risk, not diagnosis" wording, a clear referral path, clinician-delivered results in pilots
MESA access is delayed	Medium	Low	Use the open PhysioNet 2018 set
Someone has already done it	Low	High	Novelty check in week 1; narrow the claim to what is new

Risk	Likelihood (our estimate)	Impact	Mitigation
Health data is leaked or misused	Low	High	No audio stored, encryption, opt-in uploads, no health data used for ads

How this proposal maps to the Innovation Brief

Each of the brief's 10 questions draws on specific sections of this proposal; the word limits add up to exactly 3,000.

#	Brief question	Words	Draw from
1	Elevator Pitch	150	At a glance; What NightEcho adds
2	Team	150	Team roles, plus your own story of why you chose this problem
3	Opportunity	300	Problem and significance
4	Key Metrics (design, technology, what's new, impact, protection)	750	System overview; Sonar sensing design; What NightEcho adds; the patent step in the novelty check
5	Validation / Progress	450	Validation plan and targets; results from Phases 7 and 9; clinical outreach status
6	Market	300	Business plan: Market
7	Competition	300	Prior work table; Competitors and positioning
8	Go-To-Market	150	Go-to-market steps
9	Business Model	300	Revenue streams; Budget
10	Fundraising	150	Budget and funding sources

Judges score Innovation (30%), Storytelling and Professionalism (20%), Practicality (20%), Marketing Strategy (20%) and Finances (10%), according to the 2026–27 Student Guide.

Sources

Clinical evidence:

- Postuma et al., Risk and predictors of dementia and parkinsonism in idiopathic RBD, Brain 2019
- Haba-Rubio et al., Prevalence of RBD in the general population (HypnoLaus), SLEEP
- Skjærbæk et al., Self-applied mastoid ExG for automated RBD detection, medRxiv 2026
- Cesari et al., Isolated RBD: is screening possible?, J Sleep Res 2025
- Alzheimer's Association, 2026 Facts and Figures
- Rowe et al., Reducing dangerous nighttime events in persons with dementia, Alzheimer's & Dementia 2009
- Lim et al., Sleep fragmentation and risk of incident Alzheimer's disease, Sleep 2013
- HHS: CDC autism prevalence of 1 in 31 (ADDM 2022 data)
- Rapp et al., Actigraphy feasibility in autistic preschoolers, Sleep 2025 abstract

Prior work:

- Contactless home monitoring of Alzheimer's-related night-time behavior, Alzheimer's & Dementia 2025
- MinderCare Radar, Imperial College London
- UW CoMotion: SleepScore app inspired by ApneaApp research
- Detection of sleep apnea using sonar smartphone technology, IEEE EMBC 2019
- ActiTect: actigraphy-based RBD screening, npj Digital Medicine
- Detecting isolated RBD at home with a lower-back wearable, npj Digital Medicine

Datasets and equipment:

- CAP Sleep Database, PhysioNet
- MESA, National Sleep Research Resource
- You Snooze You Win: PhysioNet/CinC Challenge 2018
- Vernier Go Direct Respiration Belt

Rules and regulation:

- Conrad Challenge 2026–27 Student Guide
- ISEF rules: human participants
- FTC Health Breach Notification Rule, Federal Register 2024